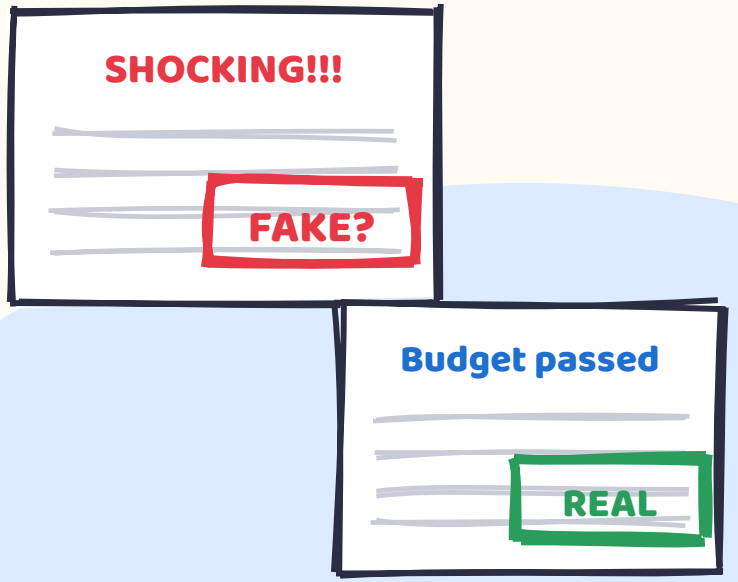
Tech Leaders · Class 11

AI Fake News Detector

Train a model, then try to break it

Datasets · TF-IDF · Naive Bayes · Media literacy



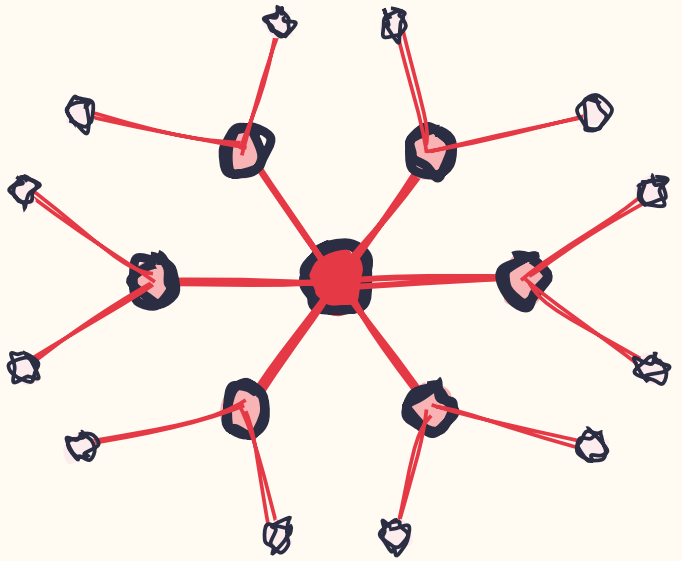
Did you know?
A 2018 MIT study of Twitter found false news reached people about six times faster than true news.

Let's think...

Can a computer tell if news is true?

Forwarded many times

A WhatsApp message says a famous drink causes illness, or that schools are closing tomorrow. It has been 'forwarded many times'. Thousands share it in an hour. Platforms use AI to flag messages like this. **How** could a program possibly know?



It reads the facts?

It checks the source?

It spots patterns!

Today: you train your own detector on real headlines, measure it, then **red-team** it to find out what it really learned.

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Learning objectives

By the end of this class I can...

1	**Why it matters:** explain how false news spreads and why platforms use AI to flag it	**Proof:** describe 2 reasons false news spreads faster than true news
2	**Dataset:** build a **balanced, sourced** dataset of real and false headlines	**Proof:** a CSV of 60+ headlines, every row with a source
3	**Text → numbers:** turn text into numbers with **TF-IDF** and train a **Naive Bayes** classifier	**Proof:** our model predicts real/fake for a new headline
4	**Evaluate:** measure **precision and recall** and inspect which words the model relies on	**Proof:** our classification report and top 10 'fake' words
5	**Limits:** fool the model, then explain why AI detects **style**, not **truth**	**Proof:** one headline that tricked it, and a human fact-check step

Objective 1 · Why it matters · video

Video time!



How false news can spread

TED-Ed (Noah Tavlin) · about 4 minutes · youtu.be/cSKGa_7XJkg

Watch for…

1. What is **circular reporting**?
2. Why do false stories spread so easily on social media?
3. What can each of us do before sharing?

Scan to watch

After the video: Think of a rumour you saw online. Where did it start? Did anyone check it?

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Objective 1 · Why it matters

Why false news wins the race

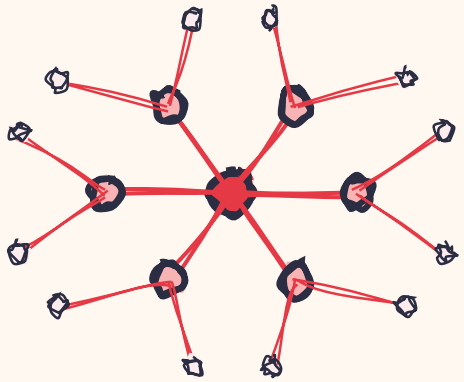
The idea

False stories are often **surprising** and **emotional** (fear, anger), so people share them faster. Platforms reward shares with more reach, so they spread further.

Humans can't check millions of posts an hour. So platforms use **AI to flag** suspicious posts for human fact-checkers. Today you build a tiny version.

Words
'False news' includes honest mistakes, satire taken seriously, and deliberate lies (disinformation).

Example: a rumour in the canteen



'The exam is cancelled!' spreads round the canteen in 5 minutes. 'The exam is still on Thursday' spreads to nobody. The exciting story wins, even though it's wrong.

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Real world

Who fights false information?



Fact-checkers

AFP Fact Check, Soch Fact Check and others verify viral claims



Platform AI

Flags posts that look like known false stories



Journalists

Confirm a story with two independent sources



You

Pause, check the source, search the claim before sharing

Which of these four is the most important line of defence? Why?

Human in the loop

AI flags. Trained humans decide. The AI alone is never the judge.

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Objective 2 · Dataset

Build a dataset you can defend

The idea

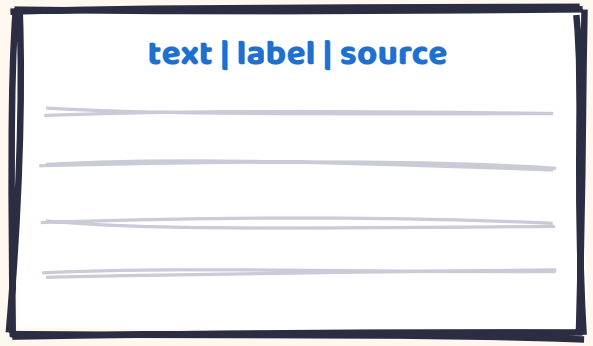
Make **headlines.csv** with columns **text, label, source**.

- **real**: headlines from established outlets (Dawn, BBC Urdu/English, APP).
- **fake**: claims rated **false** by fact-checkers (AFP Fact Check, Soch Fact Check).
- Aim for **balance**: about the same number of each.
- Every row has a **source link**. No source, no row.

Hidden bias

If all 'real' rows come from one newspaper, the model may learn that paper's writing style, not truth. That is BIAS.

Example: a fair taste test



To test which mango is sweeter, you taste the same number of each, from different trees. If all your 'good' mangoes came from one tree, you'd learn about that tree, not about mangoes.

Objective 3 · Text → numbers

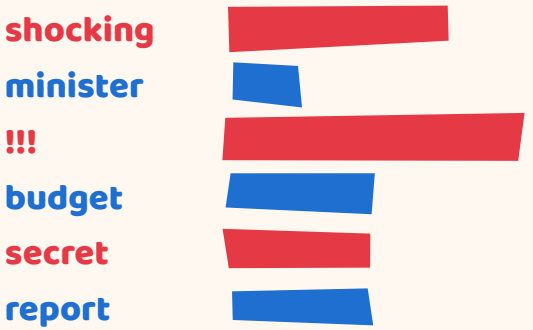
Turning words into numbers

The idea

Computers can't read. **Bag of words** counts each word. **TF-IDF** goes further: words that are common in one headline but **rare overall** get a high score ('shocking'), while words everywhere ('the', 'is') score low.

Each headline becomes a long list of numbers, one per word in our vocabulary.

Example: a highlighter



When you revise, you don't highlight 'the' and 'and'. You highlight the rare, important words. TF-IDF is an automatic highlighter.

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Objective 3 · Text → numbers

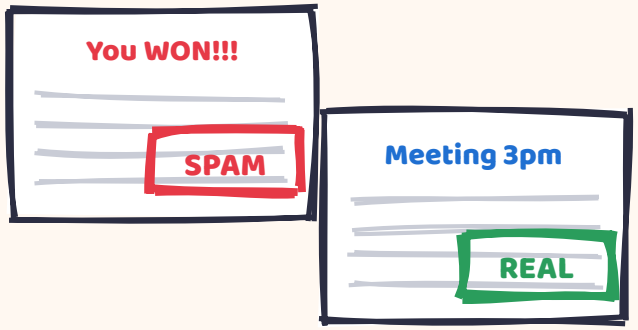
Naive Bayes: which pile is this word from?

The idea

During training the model counts how often each word appears in **real** vs **fake** headlines. For a new headline, it multiplies up the evidence from every word and picks the more likely label.

'Naive' because it treats each word as independent, ignoring word order. It is simple, fast and works surprisingly well on text.

Example: spam filters



Early email spam filters used Naive Bayes. Words like 'winner', 'free' and 'urgent' pushed an email towards the spam pile. Our detector does the same with headlines.

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Objective 4 · Evaluate

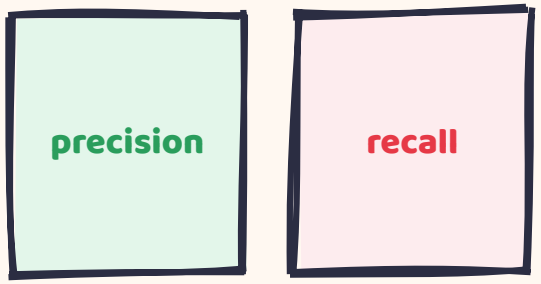
Precision and recall for 'fake'

The idea

Precision: of the headlines we flagged as fake, how many really were? Low precision = we wrongly accuse real news.
Recall: of all the fake headlines, how many did we catch? Low recall = fakes slip through.

A platform flagging posts for humans may want **high recall**. A system that deletes posts automatically needs **very high precision**.

Example: a school security guard



A guard who stops everyone catches every intruder (high recall) but annoys every student (low precision). A guard who stops nobody never bothers students but catches nobody. Good systems balance both.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Real or fake?
2. Train the detector
3. Open the black box
4. Try to fool your model
5. SIFT: four moves before you share
6. Results log
7. Detector challenges

Class game

Real or fake?

How to play

1. Teacher shows 12 headlines, one at a time (from real news and verified fact-checks).
2. Each student votes **REAL** or **FAKE** with a thumb, in 5 seconds.
3. Reveal the answer and the source.
4. Keep score. Note the clues you used: CAPITALS? !!!? named source? numbers?

Then discuss

Which clues helped? Which clues could a computer also see? Which needed knowledge of the world?



Big idea

You judged mostly by STYLE. Remember that: our AI will too.

MAKE

Objective 3 · Text → numbers · Build step 1 of 2

Train the detector

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.pipeline import make_pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report

df = pd.read_csv("headlines.csv")              # text, label, source
print(df.label.value_counts())                 # is it balanced?

X_tr, X_te, y_tr, y_te = train_test_split(
    df.text, df.label, test_size=0.25, stratify=df.label, random_state=0)

model = make_pipeline(TfidfVectorizer(ngram_range=(1, 2)), MultinomialNB())
model.fit(X_tr, y_tr)
print(classification_report(y_te, model.predict(X_te)))

while True:
    h = input("Headline: ")
    print(model.predict([h])[0], round(model.predict_proba([h]).max(), 2))
```

pipeline

TF-IDF and the classifier glued together. New text goes through both steps automatically.

stratify

Keeps the same real/fake balance in training and testing.

ngram (1, 2)

Also counts word **pairs** like 'prime minister', which carry more meaning.

MAKE

Objective 4 · Evaluate · Build step 2 of 2

Open the black box

```
import numpy as np
vec = model.named_steps["tfidfvectorizer"]
nb  = model.named_steps["multinomialnb"]
words = vec.get_feature_names_out()

print(nb.classes_)                    # ['fake' 'real'] (alphabetical)
lean = nb.feature_log_prob_[0] - nb.feature_log_prob_[1]
order = np.argsort(lean)

print("Most 'fake' words:", words[order[-12:]][::-1])
print("Most 'real' words:", words[order[:12]])
```

Read the lists

Are the 'fake' words about **truth** (facts that are wrong) or about **style** (shocking, viral, !!!)?

Source leak?

If 'dawn' or 'bbc' appears as a 'real' word, the model is learning the **outlet name**, not the content. Remove those from the text and retrain.

Explainability

Being able to see WHY a model decided is a big deal in real AI systems.

MAKE

Objective 5 · Limits · red team

Try to fool your model

Attack	Model said	Fooled?
A TRUE headline written in a shocking style		
A FALSE claim written calmly, like a news agency		
A real headline with 'minister' swapped for 'aliens'		
Satire (a joke article)		
A headline in Urdu or Roman Urdu		

What you'll find

The model judges **how** something is written, not **whether it is true**. A calm lie passes. A shocking truth gets flagged.

This is why

Platforms send flagged posts to HUMAN fact-checkers, who check sources and evidence.

Be the human in the loop

SIFT: four moves before you share

Stop. Strong emotion? That's your cue to slow down.

Investigate the source. Who published it? Are they real and accountable?

Find better coverage. Search the claim. Do trusted outlets or fact-checkers report it?

Trace it back. Find the original photo, quote or study. Is it from a different year or place?



SIFT

A method from digital literacy educator Mike Caulfield. It takes about 60 seconds.

MAKE

Test like an engineer

Results log

Measure	Ours	Notes
Dataset size / real : fake		
Precision (fake)		
Recall (fake)		
Top 3 'fake' words		
Attacks that fooled it		/ 5

Debug checklist

KeyError 'label'? Check the CSV header spelling.
Every prediction the same? Dataset too small or unbalanced: add rows.

With only 60 rows, results jump around. Run with random_state = 0, 1, 2 and report the range, not your best run.

MAKE

Level up!

Detector challenges

Choose your challenge

Compare models: swap MultinomialNB for LogisticRegression. Better precision?

Bigger data: use a larger public dataset your teacher provides. Does it generalise to Pakistani news?

Remove the leak: strip outlet names and retrain. Did accuracy drop? Why is that GOOD?

Web app: wrap the model in a simple Streamlit or Flask page.

Think like an engineer

If someone knew exactly which words your model flags, how would they write fake news to avoid it?

Boss level

Write a 1-page 'model card': data, metrics, attacks that worked, and who should NOT rely on it.

Debate

Should AI automatically delete 'fake' posts?

For

Lies spread in minutes. Humans are too slow. Automatic removal protects people during emergencies and elections.

Against

Our model flagged true news written in an exciting style. Who decides what is 'true'? Automatic removal could silence real reporting or minority voices.

Vote, then debate for 6 minutes. Can you agree on a **middle ground**, like warning labels or slowing down sharing?

Objective check

Can I really do it? Prove it.

01	Why does false news often spread faster than true news?	○ ○ ○
02	Show your CSV. How did you make it balanced, and where is each source?	○ ○ ○
03	Explain TF-IDF in one sentence. Then show the model predicting a new headline.	○ ○ ○
04	Read your precision and recall for 'fake'. Which matters more for a warning label?	○ ○ ○
05	Show a headline that fooled your model and explain why it worked.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Red-team swap: try to fool another team's model. The team that is hardest to fool presents its dataset.

Show what you know

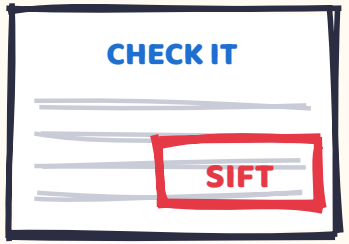
What did we learn?

- [x] False news spreads fast because it is surprising and emotional
- [x] A model is only as good and fair as its dataset
- [x] TF-IDF turns text into numbers; Naive Bayes weighs word evidence
- [x] Precision and recall show different kinds of mistakes
- [x] AI detects style, not truth: humans must check sources

Exit ticket: your model flags a true headline as fake. Name two possible reasons.

At home

Before sharing anything this week, use SIFT once. Tell us what you found.



Next class: Web-Controlled Devices via REST API!